Project 2: YSU Quantitative Finance

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Outline

Bayesian Updates: Bernoulli Data

Model each binary outcome $X_i \in \{0, 1\}$ (e.g. is the daily return positive?) as $X_i \overset{iid}{\sim} \text{Bernoulli}(p)$.
Prior: $p \sim \text{Beta}(a, b)$, default $a = b = 1$ (Uniform)
Posterior (conjugacy), with $s = \sum x_i$ successes out of n:

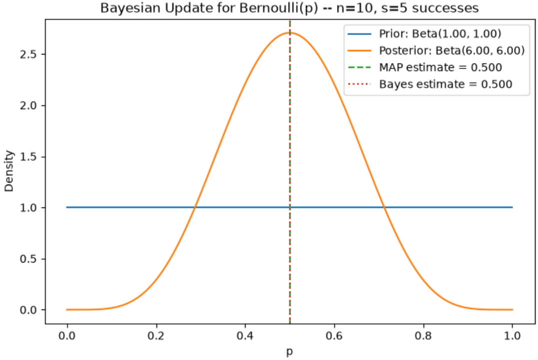
$$p \mid \text{data} \sim \text{Beta}(a + s,\ \ b + n - s)$$

Estimates from posterior $\text{Beta}(a', b')$:

$$\hat{p}_{\text{MAP}} = \frac{a' - 1}{a' + b' - 2}, \qquad \hat{p}_{\text{Bayes}} = \frac{a'}{a' + b'}$$

Bernoulli Example: Toy Data

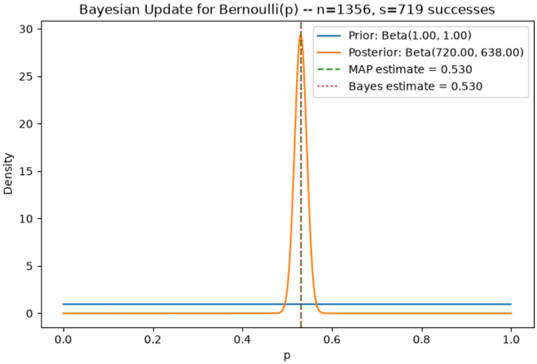
Prior Beta$(1, 1)$, data 1100101100 ($n = 10$, $s = 5$ successes)



Posterior Beta$(6, 6)$, $\quad \hat{p}_{\text{MAP}} = \hat{p}_{\text{Bayes}} = 0.500$

Bernoulli Example: Is AAPL's Daily Return Positive?

$X_t = 1$ if AAPL's daily log return is positive, 0 otherwise. Prior Beta$(1, 1)$, $n = 1356$ days, $s = 719$ positive days.



Posterior Beta$(720, 638)$, $\quad \hat{p}_{\text{MAP}} = \hat{p}_{\text{Bayes}} \approx 0.530$

- AAPL closes up on $\approx 53\%$ of days – slightly more often than not

Bayesian Updates: Normal Data

Model $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$ with **known** σ (e.g. the mean daily return μ of a stock).
Prior: $\mu \sim N(\mu_0, \sigma_0^2)$, default $\mu_0 = 0$, $\sigma_0 = 1$
Posterior (conjugacy), with sample mean $\bar{x}$:

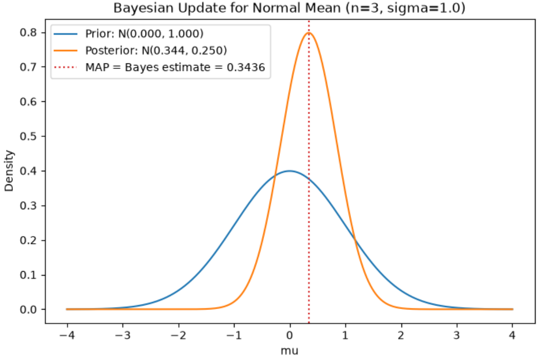
$$\mu \mid \text{data} \sim N(\mu_n, \sigma_n^2), \qquad \frac{1}{\sigma_n^2} = \frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}, \qquad \mu_n = \sigma_n^2 \left(\frac{\mu_0}{\sigma_0^2} + \frac{n\bar{x}}{\sigma^2} \right)$$

For a Normal posterior, mode = mean:

$$\hat{\mu}_{\text{MAP}} = \hat{\mu}_{\text{Bayes}} = \mu_n$$

Normal Example: Toy Data

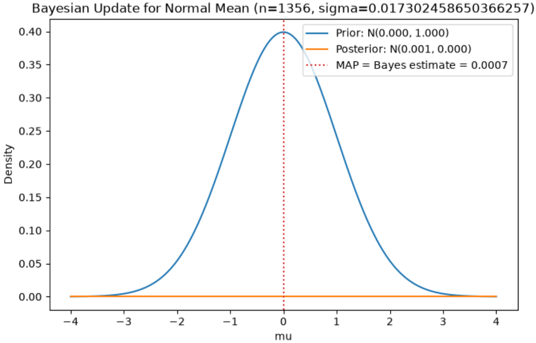
$\sigma = 1$, prior $N(0, 1)$, data $0.123,\ 1.3812,\ -0.13\ (n = 3)$



Posterior $N(0.344,\ 0.25)$, $\quad \hat{\mu}_{\text{MAP}} = \hat{\mu}_{\text{Bayes}} = 0.344$

Normal Example: Mean of AAPL Daily Returns

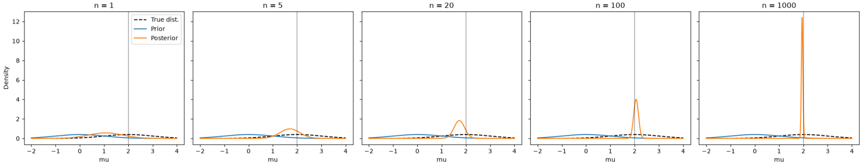
σ fixed at the sample std of AAPL's log returns, prior $N(0, 1)$, $n = 1356$



Posterior mean $\hat{\mu}_{\text{Bayes}} \approx 0.0007$ – with $n = 1356$, the data completely dominates the (wide) prior

Supplementary: Does the Posterior Forget the Prior?

Known true $\mu = 2$, $\sigma = 1$; prior $N(0, 1)$ (deliberately off-center). Generate n points from the true distribution and compute the posterior.



- As n grows, the posterior narrows and shifts toward the true μ
- The prior's influence vanishes as $n \to \infty$

Data and Methodology

- **Tickers:** `^IXIC` (NASDAQ Composite), `AAPL` (Apple), `GOOGL` (Alphabet)
- **Period:** 2021-01-01 to 2026-06-01, daily data ($\approx$ 1356 observations)
- **Variable:** daily log returns

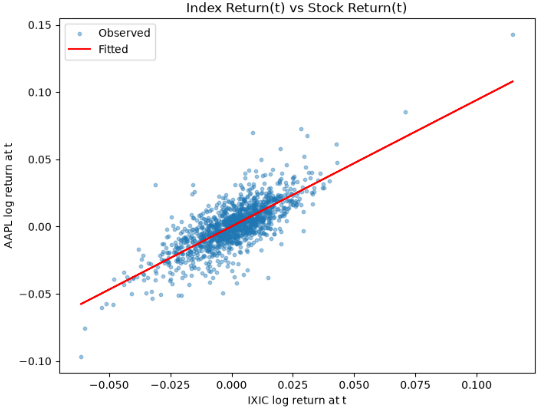
$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right)$$

- **Method:** Ordinary Least Squares (OLS), via `statsmodels`

Model 1: Index Return(t) $\to$ Stock Return(t)

Question: Can market index returns explain individual stock returns?

$$\text{AAPL Return}_t = \beta_0 + \beta_1 \cdot (\text{NASDAQ Return})_t + \varepsilon_t$$



Model 1: Results

$$\widehat{\text{AAPL}}_t = 0.00015 + 0.940 \cdot \text{NASDAQ}_t$$

Coefficient	Estimate	p-value
β_0 (intercept)	0.00015	0.625 (not significant)
β_1 (NASDAQ_t)	0.940	$\approx 5 \times 10^{-260}$

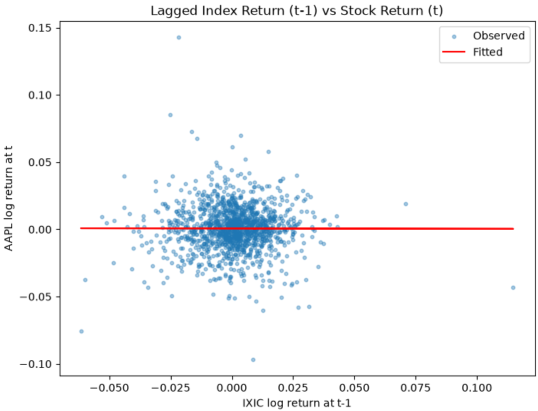
$R^2 = 0.584, \quad \text{Adjusted } R^2 = 0.583$

- AAPL's daily return moves almost one-for-one with the NASDAQ ($\beta_1 \approx 1$)
- $\approx 58\%$ of AAPL's return variance is explained by the index return
- Residuals are fat-tailed (kurtosis ≈ 6) – typical of daily financial returns

Model 2: Lagged Index Return($t-1$) $\rightarrow$ Stock Return(t)

Question: Can past returns help predict future returns?

$$\text{AAPL Return}_t = \beta_0 + \beta_1 \cdot (\text{NASDAQ Return})_{t-1} + \varepsilon_t$$



Model 2: Results

$$\widehat{\mathrm{AAPL}}_t = 0.00066 - 0.0024 \cdot \mathrm{NASDAQ}_{t-1}$$

Coefficient	Estimate	p-value
β_0 (intercept)	0.00066	0.159 (not significant)
β_1 (NASDAQ_{t-1})	-0.0024	0.943 (not significant)

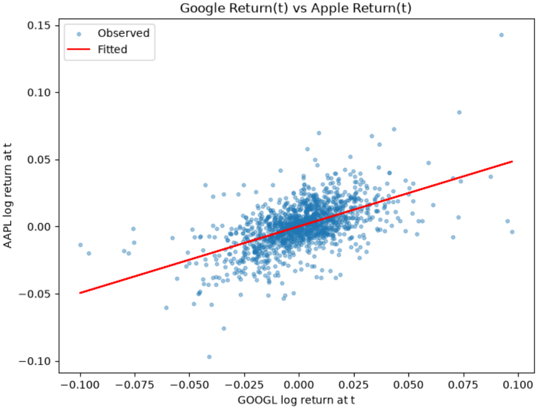
$R^2 \approx 0.0000$, Adjusted $R^2 \approx -0.0007$

- Yesterday's NASDAQ return has **no** predictive power for today's AAPL return
- Consistent with the (weak-form) Efficient Market Hypothesis

Model 3: Google Return(t) $\rightarrow$ Apple Return(t)

Question: Can one stock's returns explain another stock's returns?

$$\text{AAPL Return}_t = \beta_0 + \beta_1 \cdot (\text{GOOGL Return})_t + \varepsilon_t$$



Model 3: Results

$$\widehat{\text{AAPL}}_t = 0.00013 + 0.496 \cdot \text{GOOGL}_t$$

Coefficient	Estimate	p-value
β_0 (intercept)	0.00013	0.750 (not significant)
β_1 (GOOGL_t)	0.496	$\approx 6.5 \times 10^{-113}$

$R^2 = 0.314$, Adjusted $R^2 = 0.313$

- A 1% same-day move in GOOGL is associated with a $\approx$ 0.5% move in AAPL
- $\approx$ 31% of AAPL's variance explained – less than the index (Model 1, 58%)
- Suggests a common market/sector factor, not a direct causal link

Stage 1: Summary

Model	Predictor	R^2	Significant?
1	NASDAQ_t	0.584	Yes
2	NASDAQ_{t-1}	0.000	No
3	GOOGL_t	0.314	Yes

- Same-day market index return strongly explains AAPL's return ($R^2 = 0.58$)
- Lagged (past) returns carry **no** predictive information – market efficiency
- Same-day peer-stock (GOOGL) return also explains a sizeable share ($R^2 = 0.31$), likely via shared exposure to the market factor

Model A: Full Lagged Model

Question: Can past returns predict future returns?

$$\text{AAPL}_t = \beta_0 + \beta_1 \text{NASDAQ}_{t-1} + \beta_2 \text{AAPL}_{t-1} +$$

$$+\beta_3 \text{MA}_{20} + \beta_4 \text{MA}_{50} + \beta_5 \text{MA}_{200} +$$

$$+\beta_6 \text{VOL}_{20} + \beta_7 \text{VOL}_{20} + \beta_8 \text{VOL}_{20} +$$

$$+\beta_9 \Delta\text{Volume}_{t-1} + \beta_{10} \text{VolRatio}_{t-1} + \varepsilon_t$$

$R^2 = 0.010, \quad$ Adjusted $R^2 = 0.001, \quad$ F-test $p = 0.336$

- Overall model is **not statistically significant**
- Only `VolumeChg_lag1` is individually significant ($p = 0.031$)

Model B: Reduced Model

$$\text{AAPL}_t = \beta_0 + \beta_1 \cdot \text{NASDAQ}_{t-1} + \beta_2 \cdot \text{AAPL}_{t-1} + \varepsilon_t$$

Coefficient	Estimate	p-value
β_0 (intercept)	0.0007	0.201
β_1 (`NASDAQ_lag1`)	-0.0007	0.383
β_2 (`AAPL_lag1`)	0.0009	0.243

$R^2 = 0.001$, Adjusted $R^2 = -0.001$, F-test $p = 0.506$

- Mirrors Stage 1 Model 2: lagged index return still has zero predictive power
- Own lagged return adds nothing either

Model C: Volume-Focused Model

Question: Can trading volume explain/predict returns?

$$\text{AAPL}_t = \beta_0 + \beta_1 \cdot \text{AAPL}_{t-1} + \beta_2 \cdot \Delta\text{Volume}_{t-1} + \beta_3 \cdot \text{VolRatio}_{t-1} + \varepsilon_t$$

Coefficient	Estimate	p-value
β_0 (intercept)	0.0007	0.200
β_1 (`AAPL_lag1`)	0.0004	0.471
β_2 (`VolumeChg_lag1`)	-0.0013	0.032
β_3 (`VolumeRatio_lag1`)	0.0009	0.158

$R^2 = 0.005$, Adjusted $R^2 = 0.002$, F-test $p = 0.152$

- Overall model not significant, but `VolumeChg_lag1` is ($p = 0.032$)
- With 10 predictors tested in Model A, one hit at $p \approx 0.03$ is roughly what chance alone would produce

Model Comparison

Model	R^2	Adj. R^2	AIC	BIC
A (full lagged)	0.010	0.001	-6053.4	-5997.9
B (reduced)	0.001	-0.001	-6059.5	-6044.3
C (volume-focused)	0.005	0.002	-6061.4	-6041.2

- Model C has the lowest AIC; Model B has the lowest BIC – the differences are negligible
- Model A is worst on both criteria (most parameters, no gain in fit)
- **None** of the three models pass the overall F-test at the 5% level

Residual Diagnostics (Model C)

- No significant autocorrelation in residuals
- Residuals are fat-tailed (kurtosis ≈ 8.7) – normality is rejected
- Same pattern as Stage 1 – typical of daily financial returns, and largely independent of which (weak) model is used

Stage 2/3: Summary

- **Market index returns:** explain AAPL *contemporaneously* ($R^2 = 0.58$, Stage 1), but the lagged index has **zero** predictive power ($R^2 \approx 0$)
- **Past returns:** across Models A, B, C, $R^2 \leq 1\%$ and the F-test is never significant – past returns do **not** predict future returns
- **Trading volume:** `VolumeChg_lag1` is marginally significant alone ($p \approx 0.03$), but the overall volume model is not ($p = 0.152$) – weak, inconclusive evidence
- **Conclusion:** results are consistent with (weak-form) market efficiency – information is priced in immediately, leaving no exploitable pattern in lagged data

Defining Market Regimes

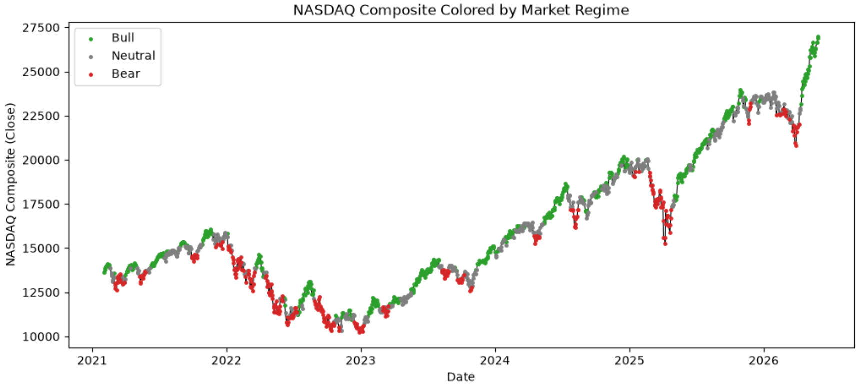
Classify each day's NASDAQ Composite (`^IXIC`) into **Bull**, **Neutral**, or **Bear** using the 20-day rolling cumulative log return

$$R_t = \sum_{i=0}^{19} r_{t-i}, \qquad \theta = 0.5 \cdot \mathrm{std}(R_t)$$

$$\mathrm{Regime}_t = \begin{cases} \text{Bull} & R_t > \theta \\ \text{Bear} & R_t < -\theta \\ \text{Neutral} & \text{otherwise} \end{cases}$$

Regime	Bull	Neutral	Bear
Days (%)	554 (41.4%)	465 (34.8%)	318 (23.8%)

NASDAQ Composite Colored by Regime



Transition Probability Matrix

$$P_{ij} = P(\text{Regime}_{t+1} = j \mid \text{Regime}_t = i)$$

$$P = \begin{pmatrix} 0.879 & 0.121 & 0.000 \\ 0.144 & 0.751 & 0.105 \\ 0.000 & 0.154 & 0.846 \end{pmatrix} \quad \text{(rows/cols: Bull, Neutral, Bear)}$$

- All three regimes are highly **persistent** (diagonal $\approx$ 0.75–0.88)
- $P(\text{Bull} \to \text{Bear}) = P(\text{Bear} \to \text{Bull}) = 0$ – the market always passes through Neutral

Stationary Distribution

$\pi P = \pi$, $\sum_i \pi_i = 1$

Regime	Stationary π	Observed frequency
Bull	0.414	0.414
Neutral	0.348	0.348
Bear	0.238	0.238

- Stationary distribution matches observed frequencies almost exactly – the chain is essentially already at its long-run distribution

Predicting the Next Regime

Given the current regime i, the most probable next regime is $\arg\max_j P_{ij}$.

- Last observed regime: **Bull**
- $P(\text{Bull} \to \text{Bull}) = 0.879, \quad P(\text{Bull} \to \text{Neutral}) = 0.121,$ $P(\text{Bull} \to \text{Bear}) = 0.000$
- **Predicted next regime: Bull**

Simulating the Markov Chain

Simulate a regime sequence of the same length from P and compare to the observed data.

	Frequency	
Regime	Observed	Simulated
Bull	0.414	0.476
Neutral	0.348	0.301
Bear	0.238	0.223

	Avg. run length (days)	
Regime	Observed	Simulated
Bull	8.1	8.8
Neutral	4.0	3.3
Bear	6.5	6.0

- A first-order Markov chain reproduces both the regime frequencies and the average regime durations reasonably well

Part III: Summary

- Market regimes (Bull/Neutral/Bear) are **highly persistent**, and the chain is essentially at its stationary distribution
- The model predicts the current Bull regime will most likely continue
- However, persistence is largely **mechanical**: R_t is a 20-day rolling sum, so R_t and R_{t-1} share 19 of 20 terms and are highly correlated by construction
- Consistent with Stage 1: *daily returns* themselves remain essentially unpredictable (weak-form efficiency); only the smoothed regime label is persistent

Questions?

Thank you!